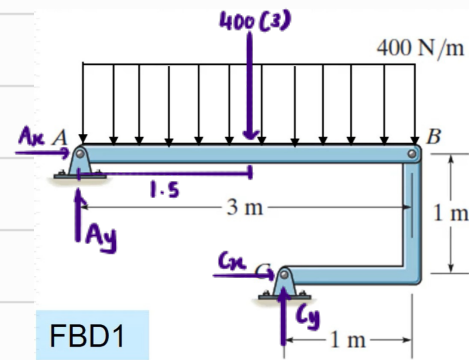
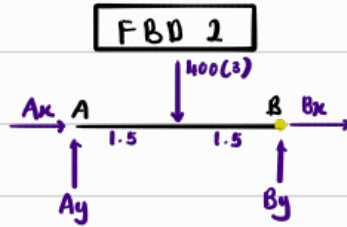


Example 1

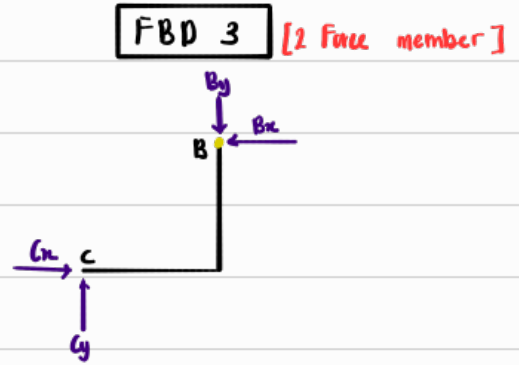
Solve for all unknown forces



FBD1



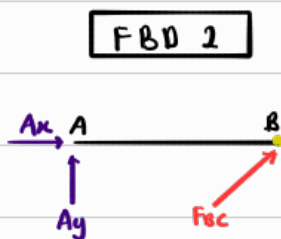
FBD 2



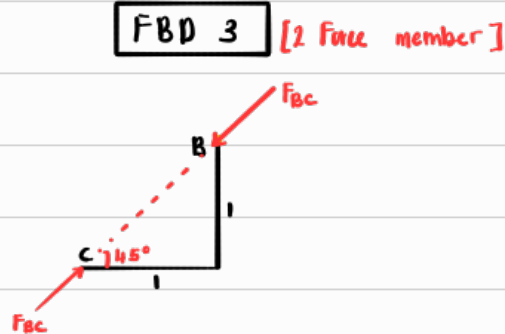
FBD 3

[2 force member]

Since FBD 3 is a 2 force member, the FBDs can be redrawn,



FBD 2



FBD 3

[2 force member]

① Looking at FBD 2,

By symmetry, $A_y = B_y$

$$\therefore A_y + B_y - 400(3) = 0$$

$$2A_y = 1200$$

$$A_y = 600 \text{ N}$$

$$\therefore B_y = 600 \text{ N}$$

② Looking at FBD 1

$$\sum F_y = 600 - 1200 + C_y$$

$$C_y = 600 \text{ N}$$

$$\sum M_A = -(1200 \times 1.5) + (600 \times 2) + C_x(1)$$

$$C_x = 600 \text{ N}$$

$$\sum F_x = A_x + 600$$

$$A_x = -600 \text{ N}$$

③ Looking back at FBD 2

$$\sum F_x = A_x + B_x$$

$$0 = -600 + B_x$$

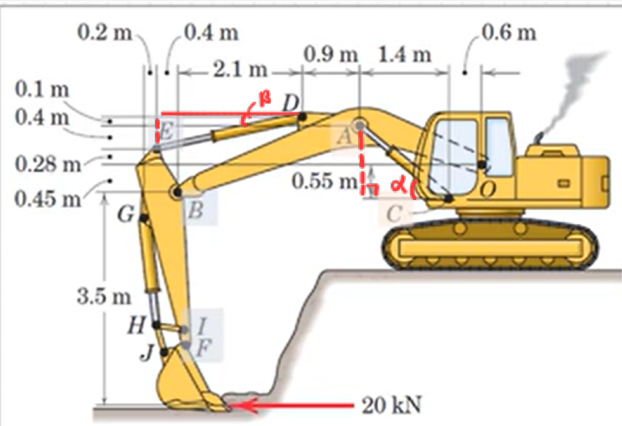
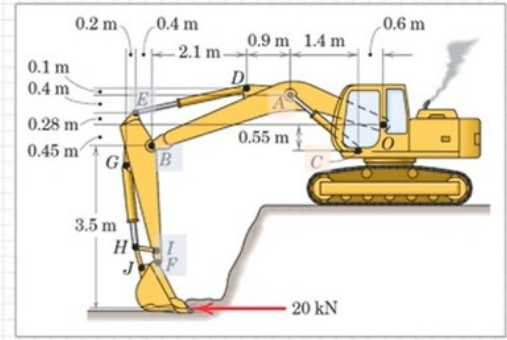
$$B_x = 600 \text{ N}$$

This method does not utilise the 2 force member, however u can if u wanted to

Example 2 :

Machine example

The excavator applies a 20 kN force parallel to the ground. There are two hydraulic cylinders at AC to control the OAB arm and a single cylinder DE to control the EBIF arm. Determine the force in the hydraulic cylinders AC & DE



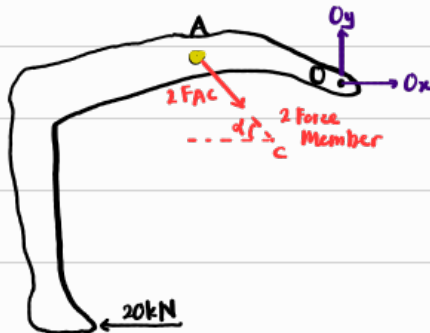
$$\tan \alpha = \frac{0.55 + 0.28 + 0.4}{1.4}$$

$$\alpha = 41.3^\circ$$

$$\tan \beta = \frac{0.1 + 0.4}{0.4 + 2.1}$$

$$\beta = 11.31^\circ$$

FBD 1



$$\sum M_O = 0$$

$$-20,000(3.95) - [2F_{AC} \cos(41.3^\circ)](0.68) + [2F_{AC} \sin(41.3^\circ)](2) = 0$$

$$F_{AC} = 48.8 \text{ kN} \#$$

FBD 2



$$\sum M_B = 0$$

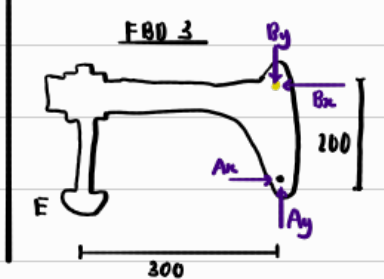
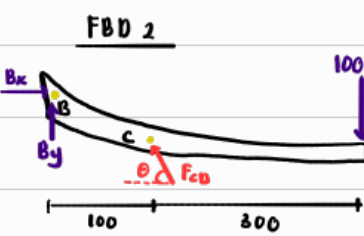
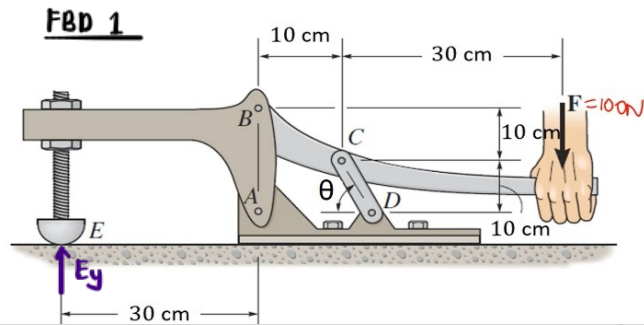
$$-20,000(3.5) + [F_{DE} \cos(11.31^\circ)](0.73) + [F_{DE} \sin(11.31^\circ)](0.4) = 0$$

$$F_{DE} = 88.1 \text{ kN} \#$$

Toggle clamp Example

A toggle clamp is a type of mechanical clamp that is commonly used in manufacturing and woodworking applications to hold workpieces in place during machining or assembly processes.

A toggle clamp is subjected to a force $F = 100 \text{ N}$ at the handle. Determine the magnitude of the vertical clamping force acting at E and the mechanical advantage of this machine.



Looking at FBD 2

$$\sum M_B = 0$$

$$-100(400) + (F_{CB} \sin \theta)(100) - (F_{CB} \cos \theta)(100) = 0$$

$$F_{cp} = \frac{400}{(\sin \theta - \cos \theta)} \quad \text{--- (1)}$$

$$\sum F_x = 0$$

$$B_x - F_{cy} \cos \theta = 0$$

$$B_{xc} = F_{cp} \cos \theta \quad \text{--- (2)}$$

$$\sum F_y = 0$$

$$B_y + F_{CD} \sin \theta - 100 = 0$$

$$B_y = 100 - F_{CD} \sin \theta \quad \text{--- (3)}$$

Looking at FBD 3

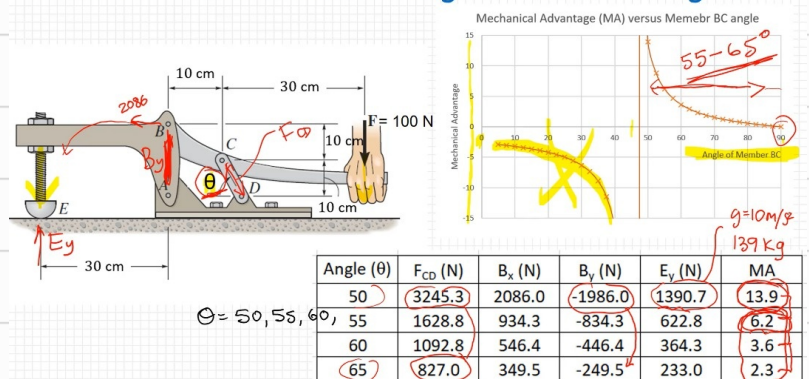
$$\bar{z} M_A = 0$$

$$(B_x)(200) - (E_y)(300) = 0$$

$$E_y = \frac{200 \text{ Bx}}{300}$$

$$F_y = \frac{800 \cos \theta}{3(\sin \theta - \cos \theta)} \quad [\text{using (1) \& (2)}]$$

Forces & Mechanical advantage for different angles

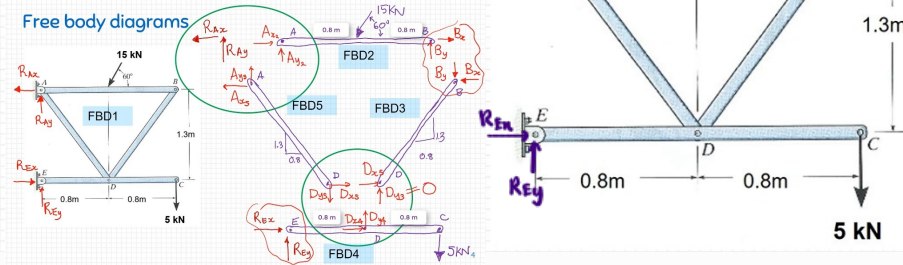


Example 4:

Four member frame

For the frame shown;

- Find the support reactions
- Calculate the axial forces in members AD and BD



① Looking at FBD 1

$$\sum M_E = 0$$

$$1.3 R_{Ax} - (12.99)(0.8) + (7.5)(1.3) - (5)(1.6) = 0$$

$$R_{Ax} = 6.6476 \text{ kN} \#$$

$$\sum F_y = 0$$

$$R_{Ay} - 12.99 - 5 + R_{Ey} = 0$$

$$R_{Ay} + R_{Ey} = 17.99 \quad \text{--- ①}$$

$$\sum F_x = 0$$

$$R_{Ex} - 6.6476 - 7.5 = 0$$

$$R_{Ex} = 14.1475 \text{ kN} \#$$

② Looking at FBD 4

$$\sum M_D = 0$$

$$-(5)(0.8) - R_{Ey}(0.8) = 0$$

$$R_{Ey} = -5 \text{ kN} \#$$

∴ Using ①,

$$R_{Ay} - 5 = 17.99$$

$$R_{Ay} = 22.99 \text{ kN} \#$$

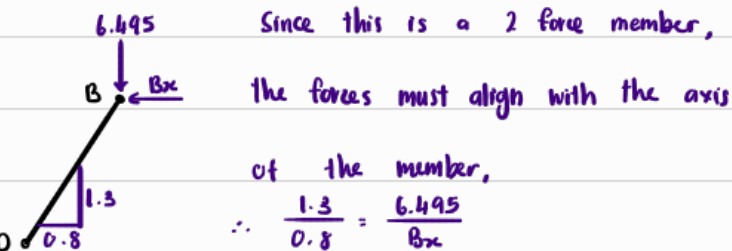
③ Looking at FBD 2

By symmetry, $A_{y2} = B_y = \frac{12.99}{2} = 6.495 \text{ kN}$

$$\sum F_x = 0$$

$$A_{x2} - 7.5 + B_x = 0 \quad \text{--- ②}$$

④ Looking at FBD 3



Since this is a 2 force member, the forces must align with the axis

of the member,

$$\therefore \frac{1.3}{0.8} = \frac{6.495}{B_x}$$

$$B_x = 4 \text{ kN} \#$$

$$\therefore F_{BD} = \sqrt{(6.495)^2 + 4^2} = 7.628 \text{ kN (compression)}$$

∴ Using ②,

$$A_{x2} - 7.5 + 4 = 0$$

$$A_{x2} = 3.5 \text{ kN}$$

Looking at FBD 5

$$\sum F_x = 6.6476$$

$$3.5 + A_{x5} = 6.6476$$

$$A_{x5} = 10.147 \text{ kN}$$

$$\sum F_y = 22.99$$

$$6.495 + A_{y5} = 22.99$$

$$A_{y5} = 16.495 \text{ kN}$$

$$\therefore F_{AD} = \sqrt{(10.147)^2 + (16.495)^2} = 19.366 \text{ kN (Tension)}$$

$$\therefore F_{BD} = 7.628 \text{ (c)}$$

$$F_{AD} = 19.366 \text{ (t)}$$

